

$$q(x) = y - 2$$
$$= y = 2.$$

$$p(2) = 4 \cdot 2^3 + a(4) - 12(2) + 6.$$

$$B = 8 + 4a - 24 + 6.$$

$$B = 4a - 10 \quad \text{--- (2)}$$

From (1) & (2),

$$(5a - 6) \times 2 = 4a - 10.$$

$$10a - 12 = 4a - 10$$

$$6a = 2$$

$$a = \frac{2}{6}$$

$$a = 1$$

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$$\angle CED = 180 - x \quad [\text{supp. } \angle\text{s}]$$

$$180 - x + 50 = 110 \quad [\text{Ext. } \angle\text{s}]$$

$$230 - 110 = x$$

$$\boxed{x = 120^\circ} //$$

$$\begin{aligned} 29. a) \quad 5^{x-2} \times 3^{2x-3} &= 135 \\ 5^{x-2} \times 3^{2x-3} &= 27 \times 5 \\ &= 3^3 \times 5^1 \end{aligned}$$

~~$$x-2+2x+3 = 3+1$$~~

~~$$3x+1 = 3+1$$~~

~~$$x=1$$~~

~~$$(x-2)(2x-3)$$~~

~~$$x-2+2x-3 = 3+1$$~~

~~$$3x = 4+5$$~~

~~$$\boxed{x = 3} //$$~~

$$b) \quad \frac{3^{2x} \cdot 3^x}{3^x} = (3^{20})^{1/4}$$

$$3^{2x} = 3^5$$

$$2x = 5$$

$$\boxed{x = \frac{5}{2}} //$$

$$20. \quad g(x) = y+1, \quad y = -1$$

$$p(-1) = (-1)^3 + 2(-1)^2 - 5a(-1) - 7$$

$$A = -1 + 2 + 5a - 7$$

$$A = 5a - 6 \quad \text{--- (1)}$$

$$6a + 2b = 22$$

$$-4a - 2b = 14$$

$$2a = 36$$

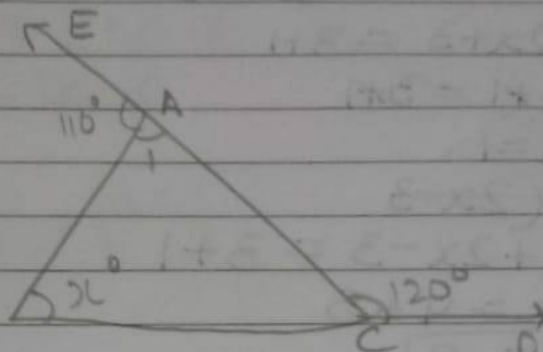
$$\boxed{a = 18}$$

$$b = 11 - 3a$$

$$b = 11 - 54$$

$$\boxed{b = -43}$$

28. a)



$$\angle 1 = 180 - 110 \quad [\text{Supp. } \angle \text{es}]$$

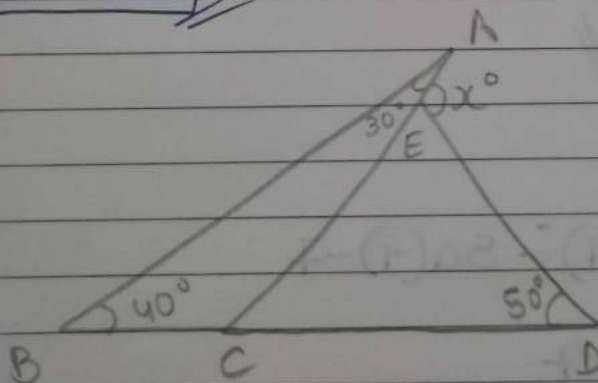
$$\angle 1 = 70^\circ$$

$$\angle 1 + x = \angle ACD \quad [\because \text{Ext. angle property}]$$

$$70^\circ + x = 120$$

$$\boxed{x = 50^\circ}$$

b)



$$\angle ACB = 180^\circ - (40 + 30) \quad [\because \text{ASP}]$$

$$= 180^\circ - 70^\circ$$

$$\angle ACB = 110^\circ$$

ii) Height corresponding to longest side.

$$h = \frac{2 \times 864}{6030} \times 28.8 = 28.8 \text{ cm}$$

26. In $\triangle ABD$ and $\triangle CBD$,

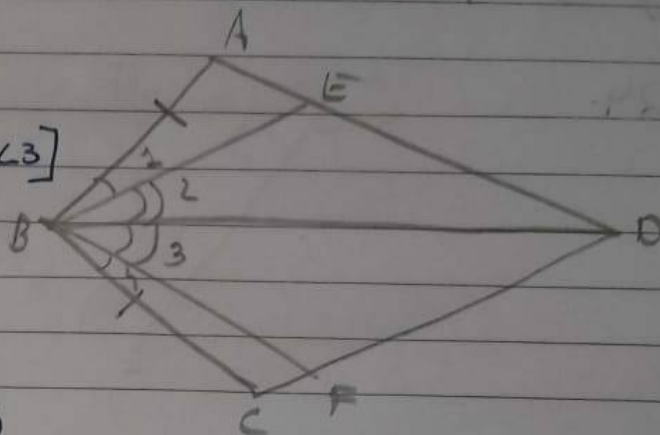
$$BD = BD \text{ (common)}$$

$$\angle 1 + \angle 2 = \angle 3 + \angle 4 \quad [\because \angle 1 = \angle 4, \angle 2 = \angle 3]$$

$$AB = BC \text{ [Given]}$$

$$\Rightarrow \triangle ABD \cong \triangle CBD \text{ [SAS]}$$

$$\angle ADB = \angle CDB \text{ [CPCT]}.$$



In $\triangle BED$ and $\triangle FCD$

$$BD = BD \text{ [common]}$$

$$\angle 2 = \angle 3 \text{ [Given]}.$$

$$\angle ADB = \angle CDB \text{ [proved]}.$$

$$\Rightarrow \triangle BED \cong \triangle FCD \text{ [ASA]}.$$

$$\therefore \underline{BE = CF} \text{ [CPCT]}.$$

Hence, proved.

27. $p(x) = x^3 + ax^2 + bx + 6.$

$$g(x) = x - 2 = 0 \Rightarrow x = 2$$

$$g(x) = x - 3 = 0 \Rightarrow x = 3$$

$$p(2) = 0$$

$$2^3 + a(2)^2 + b(2) + 6 = 0$$

$$4a + 2b = -6 - 8$$

$$4a + 2b = -14 \quad \text{--- (1)}$$

$$p(3) = 0$$

$$3^3 + 9a + 3b + 6 = 0$$

$$27 + 9a + 3b + 6 = 0$$

$$9a + 3b = -33$$

$$3a + b = -11 \quad \text{--- (2)}$$

$$6a + 2b = -22 \quad \text{--- (2)}$$

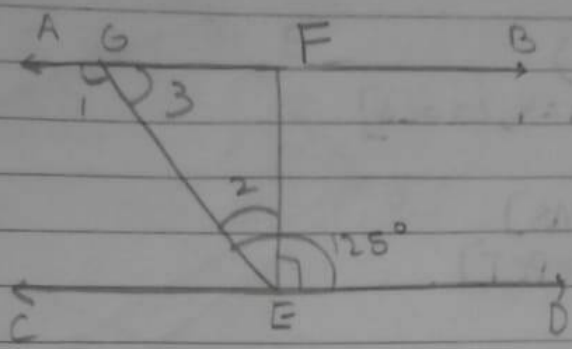
$$\begin{array}{r} 236 \\ \times 24 \\ \hline 143 \\ 720 \\ \hline 864 \end{array}$$

$$= \frac{1}{2} + \frac{11}{2} - 3$$

$$\frac{12}{2} - 3$$

$$p(\frac{1}{2}) = 3 \quad ; \text{ Remainder} = 3$$

24.



$$\angle AGE = \angle GED \quad [\text{alt. int.}]$$

$$\boxed{\angle AGE = 125^\circ}$$

$$\angle FGE = 180^\circ - \angle AGE \quad [\text{Supp. Angs.}]$$

$$\angle FGE = 180 - 125$$

$$\boxed{\angle FGE = 55^\circ}$$

$$\angle EFG = 90^\circ \quad [\text{alt. int.}]$$

$$\Rightarrow \angle 2 + \angle 3 = 180^\circ$$

$$55 + \angle 3 = 180^\circ$$

$$\boxed{\angle 3 = 125^\circ}$$

25.

$$3x + 4x + 5x = 144$$

$$12x = 144$$

$$\boxed{x = 12}$$

$$\text{Sides} = 36, 48, 60 \quad S = \frac{36 + 48 + 60}{2} = 72$$

$$\text{Area} = \sqrt{72(36)(24)(12)}$$

$$= \sqrt{864(36)(24)} = \sqrt{864 \times 864} = 864 \text{ cm}^2$$

22. a) $(100+2)^3$

$$(100+2)^3 = 100^3 + 2^3 + 3(100)(2)[100+2]$$

$$= 1000000 + 8 + \cancel{600} (60000 + 1200)$$

$$= \cancel{1072008} 1060008 + 1200$$

$$= \underline{\underline{1061208}}$$

b) $(1000-3)^2 = \cancel{1000}^2 - \cancel{3}^2$

$$= 1000^2 + 3^2 - 2(1000)(3)$$

$$= 1000000 + 9 - 6000$$

$$= 994000 + 9$$

$$= \underline{\underline{994009}}$$

23. b) i) $g(x) = 0$

$$x-2=0$$

$$x=2$$

$$p(2) = 3(2)^4 - 6(2)^2 - 8(2) - 2$$

$$= 16 \times 3 - 6 \times 4 - 16 - 2$$

$$= 48 - 24 - 18$$

$$p(2) = 6$$

$$\text{Remainder} = 6$$

(ii) $g(x) = 0$

$$\frac{x+1}{2} = 0$$

$$x = -\frac{1}{2}$$

$$p\left(-\frac{1}{2}\right) = 2\left(-\frac{1}{2}\right)^3 + 3\left(-\frac{1}{2}\right)^2 - 11\left(-\frac{1}{2}\right) - 3$$

$$= -\frac{21}{84} + \frac{3}{4} + \frac{11}{2} - 3$$

19. Area = $\sqrt{s(s-a)(s-b)(s-c)}$

$a=b=c=10\text{cm.}$

Perimeter = 30cm.

$s = 15\text{cm}$

$= \sqrt{15(5)(5)(5)}$

$= \sqrt{125 \times 15}$

$= \sqrt{1875}$

Area = $25\sqrt{3}\text{ cm}^2$

20. a) $x^2 - 4x - 21$

$x^2 - 7x + 3x - 21$

$x(x-7) + 3(x-7)$

$= (x+3)(x-7)$

b) $x^2 + 5\sqrt{3}x + 12$

$x^2 + 4\sqrt{3}x + \sqrt{3}x + 12$

$x(x+4\sqrt{3}) + \sqrt{3}(x+4\sqrt{3})$

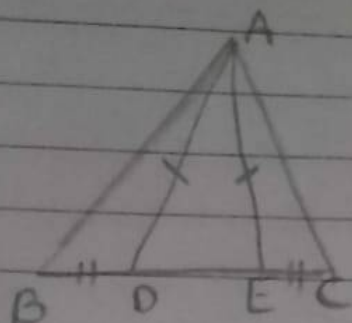
$= (x+\sqrt{3})(x+4\sqrt{3})$

21. $\left(\frac{16}{81}\right)^{+3/4} \times \left[\left(\frac{25}{9}\right)^{-3/2} \times \left(\frac{5}{2}\right)^3\right]$

$\left(\frac{2^4}{3^4}\right)^{3/4} \times \left[\left(\frac{5}{3}\right)^{-3} \times \frac{125}{8}\right]$

$\frac{8}{27} \times \left[\frac{27}{125} \times \frac{125}{8}\right] = 1 //$

17.



~~AAA~~ $\angle ADE = \angle AED$ [$\because$ $\angle$ le opp. to equal sides].

Add 180° on both sides.

$$\angle ADE + 180 = \del{AA} \angle AED + 180^\circ$$

$$180^\circ - \angle AED = 180^\circ - \angle ADE$$

$$\angle AEC = \angle ADB \quad [\because \text{supp. } \angle\text{es}]$$

In $\triangle ADB$ and $\triangle AEC$,
 $AD = AE$ [Given]

$$\angle AEC = \angle ADB \quad [\text{Proved}]$$

$$BD = EC \quad [\text{Given}]$$

$$\therefore \triangle ADB \cong \triangle AEC \quad [\text{SAS}]$$

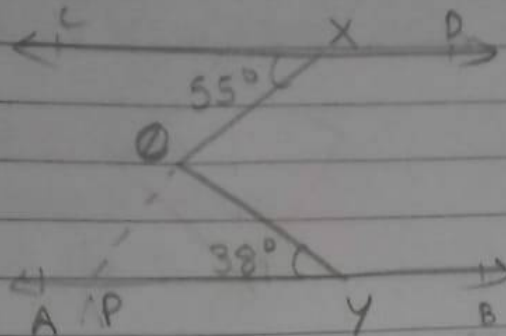
18. $\left(\frac{y}{3} - \frac{x}{2}\right)^3$

$$(a-b)^3 = a^3 - b^3 - 3ab(a-b)$$

$$\left(\frac{y}{3} - \frac{x}{2}\right)^3 = \frac{y^3}{27} - \frac{x^3}{8} - \frac{3xy}{82} \left(\frac{y}{3} - \frac{x}{2}\right)$$

$$\left(\frac{y}{3} - \frac{x}{2}\right)^3 = \frac{y^3}{27} - \frac{x^3}{8} - \frac{xy^2}{6} + \frac{x^2y}{4}$$

16)



Construction: Extend XO to meet AB at P.

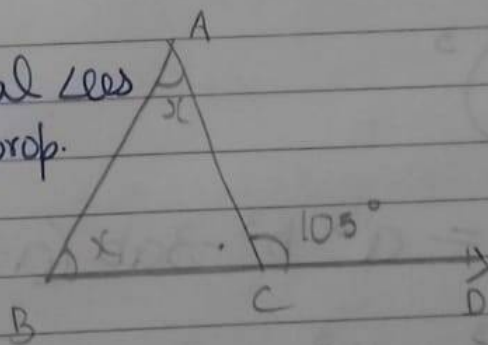
Proof: $\angle XPB = \angle PXC$ [alt. int]
 $\angle XPB = 55^\circ$

In ~~triangle~~ $\triangle YPO$,
 $\angle POY = 38^\circ + 55^\circ = 180 - (38 + 55)$ [$\because$ ASP]
 $= 180 - 93$
 $\angle POY = 87^\circ$

$\angle YOX = 180 - \angle POY$ [$\because$ supp. $\angle$ s]
 $\angle YOX = 180 - 87$
 $\angle YOX = 93^\circ$

(ii) Let x be the equal $\angle$ s
 Acc. to exterior $\angle$ s prop.

$$\begin{aligned} x + x &= 105^\circ \\ 2x &= 105^\circ \\ x &= 52.5^\circ \end{aligned}$$



14. $25^{3/2} \times 243^{3/5}$
 $16^{5/4} \times 8^{4/3}$

$$\begin{array}{r} 3 \\ 1125 \\ \times 27 \\ \hline 7875 \\ 22500 \\ \hline 30375 \end{array}$$

$$\frac{(5^2)^{3/2} \times (3^5)^{2/5}}{(2^4)^{5/4} \times (2^3)^{4/3}}$$

$$\frac{5^3 \times 3^3}{2^5 \times 2^4}$$

$$\frac{125 \times 27}{2^9}$$

$$= \frac{125 \times 27}{512} = \frac{3375}{512}$$

15. i) $\angle ABC = \angle 1 + \angle 4$
 $\angle ACB = \angle 3 + \angle 2$

$$\angle 1 + \angle 4 = \angle 3 + \angle 2 \quad [\because \text{A.T.Q.}]$$

$$\angle 1 + \angle 4 = \angle 4 + \angle 2 \quad [\because \angle 3 = \angle 4]$$

$$\boxed{\angle 1 = \angle 2}$$

Hence, proved.

(ii) Axiom: If any 2 points are given, there is a unique line that passes through them

Postulate: All right angles are equal to one another.

12. $q(x) = 3x - 2 = 0$

$$\boxed{x = \frac{2}{3}}$$

$$p\left(\frac{2}{3}\right) = 3\left(\frac{2}{3}\right)^2 - 5\left(\frac{2}{3}\right) + 2$$

$$= \frac{12}{9} - \frac{10}{3} + \frac{18}{9}$$

$$= \frac{12 - 30 + 18}{9}$$

$$= \frac{30 - 30}{9}$$

$$p\left(\frac{2}{3}\right) = 0$$

$\therefore q(x)$ is a factor of $p(x)$.

13. $2^{2x-y} = 128$, $2^{x+y} = 32$.

$$2^{2x-y} = 2^7$$

$$2x - y = 7 \text{ --- (1)}$$

$$2^{x+y} = 2^5$$

$$x + y = 5 \text{ --- (2)}$$

$$2x - y + x + y = 7 + 5$$

$$3x = 12$$

$$\boxed{x = 4} //$$

$$\boxed{y = 1} //$$

$$x^2 + y^2 = 4^2 + 1^2$$

$$\boxed{x^2 + y^2 = 17}$$

10. In $\triangle ADC$,

$$\angle DAC = \angle ADC = R \text{ [les opp. equal sides]}$$

$$R + R + 64 = 180^\circ \text{ [ASP]}$$

$$2R = 116$$

$$\boxed{R = 58^\circ}$$

$$\Rightarrow \angle AOC = 58^\circ$$

$$\angle ADB = 180 - \angle ADC \text{ [}\therefore \text{ supplementary]}$$

$$\angle ADB = 180 - 58 = 122^\circ$$

$$\boxed{\angle ADB = 122^\circ}$$

In $\triangle ADB$,

$$\angle x = \angle ABD \text{ [les opp. equal sides]}$$

By ASP,

$$\angle x + \angle ABD + \angle ADB = 180^\circ$$

$$2x + x + 122^\circ = 180$$

$$2x = 58^\circ$$

$$\boxed{x = 29^\circ}$$

11. $x + y = 12$

$$xy = 32$$

$$x^2 + y^2 = ?$$

$$(x + y)^2 = 12^2$$

$$(x + y)^2 = 144$$

$$x^2 + y^2 + 2xy = 144$$

$$x^2 + y^2 + 2(32) = 144$$

$$\boxed{x^2 + y^2 = 80} //$$

8. $P \quad R \quad S \quad Q$

$$PS = RQ.$$

$$PT: PR = SQ.$$

$$~~PS = RS~~ \quad PS = RQ$$

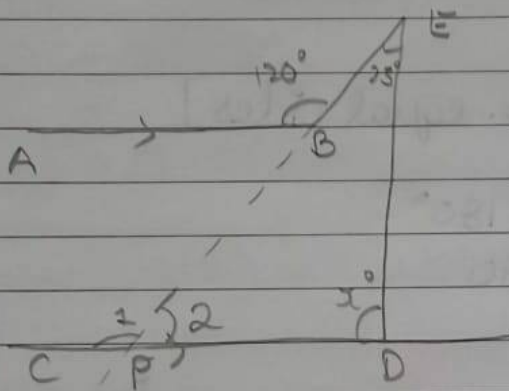
[Subtract RS on both sides].

$$PS - RS = RQ - RS.$$

$$= PR = SQ.$$

Hence, proved

9.



Construct Construction: Extends EB to CD meeting it at P.

Proof: $\angle 1 = \angle ABE$ [$\because$ corr. $\angle$ s].

$$\angle 1 = 120^\circ.$$

$$\angle 2 = 180^\circ - 120^\circ \text{ [Supplementary to } \angle 1]$$

$$\angle 2 = 60^\circ$$

In $\triangle EPD$,

$$\angle E + \angle D + \angle 2 = 180^\circ \text{ [ASP]}$$

$$25 + 60 + \angle 2 = 180$$

$$85 + \angle 2 = 180$$

$$\boxed{\angle 2 = 95^\circ}$$

5. $3y + x = 7y$
 $x = 4y$ ——— ①

$7y = 180 - 5y$ ——— ②

$12y = 180$

$y = 15^\circ$

$x = 4(15)$

$x = 60^\circ$

Option 1) 60°

6. $\frac{\sqrt{3}a^2}{4} = 36\sqrt{3}$, $a^2 = 144$
 $a = 12\text{cm}$

$a = 3$

Perimeter = 12cm

Perimeter = $3 \times 12 = 36\text{cm}$

Option 1) 36cm

7. $\frac{2 \times \sqrt{5}}{\sqrt{5} \times \sqrt{5}} = \frac{2\sqrt{5}}{5}$

$= \frac{2 \times 2.236}{5}$

$= \frac{4.472}{5} = 0.8944$

Option

Ans: 0.8944

Board Exam

1. $x = \frac{\sqrt{7}}{5}$, $\frac{5}{x} = p\sqrt{7}$

$$\frac{5}{\frac{\sqrt{7}}{5}} = p\sqrt{7}$$

$$\frac{25}{\sqrt{7}} = p\sqrt{7}$$

$$\boxed{\frac{25}{7} = p}$$

Option 2) $\frac{25}{7}$

2. $x^{100} + 2x^{99} + k$

$$p(x) = x^{100} + 2x^{99} + k.$$

$$g(x) = x + 1$$

$$\boxed{x = -1}$$

$$p(-1) = 1 + 2 + k$$

$$0 = 3 + k$$

$$-k = 3, \boxed{k = -3}$$

$$\boxed{\text{Option 1) } 1}$$

3. Let x be angle.

$$180 - x = 3(90 - x)$$

$$180 - x = 270 - 3x$$

$$2x = 90^\circ$$

$$\boxed{x = 45^\circ}$$

$$\boxed{\text{Option 4) } 45^\circ}$$